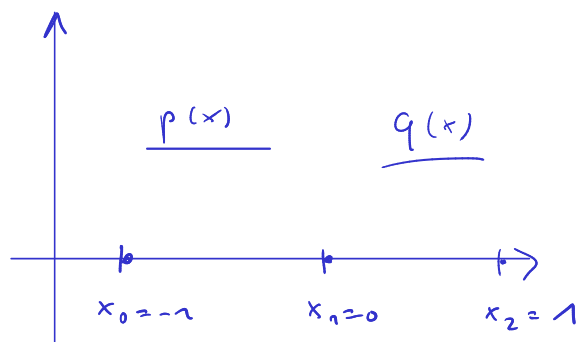


① kubischer, nat. Spline

$2n$	x_i	-1	0	1	$n=2$
	y_i	8	4	4	

direkte Methode ($4n \times 4n$ - Gl. system)



$$p(x) = a + bx + cx^2 + dx^3$$

$$q(x) = e + fx + gx^2 + hx^3$$

$$p'(x) = b + 2cx + 3dx^2$$

$$q'(x) = f + 2gx + 3hx^2$$

$$p''(x) = 2c + 6dx$$

$$q''(x) = 2g + 6hx$$

Interpolation:

$$y_0 = 8 = p(-1) = a - b + c - d \quad \times \checkmark$$

$$y_1 = 4 = p(0) = a \quad \checkmark$$

$$y_1 = 4 = q(0) = e \quad \checkmark$$

$$y_2 = 4 = q(1) = e + f + \underline{g} + h \quad \times$$

$$C^1: \quad p'(0) = q'(0) \Leftrightarrow b = f \quad \checkmark$$

$$C^2: \quad p''(0) = q''(0) \Leftrightarrow 2c = 2g \quad \checkmark$$

$$\underline{\text{Ab. Bed.}}: \quad p''(-1) = 0 = 2c - 6d \Leftrightarrow \underline{c = 3d} \quad \checkmark$$

$$q''(1) = 0 = 2g + 6h \Leftrightarrow \underline{g = -3h} \quad \checkmark$$

$$\underline{8 \times 8 \text{ LGS}}: \quad a = \underline{4} = e, \quad b = f, \quad \underline{c = g}, \quad d = -h$$

$$8 = 4 - f + 3d - d$$

$$4 = 4 + f + 3d - d$$

$$2d - f = 4$$

$$2d + f = 0$$

$$d = 1 \quad f = -2$$

$$S_3(x) = \begin{cases} p(x) & x \leq 0 \\ q(x) & x > 0 \end{cases}$$

$$p(x) = 4 - 2x + 3x^2 + x^3$$

$$q(x) = 4 - 2x + 3x^2 - x^3$$